2016.0 RANGE ROVER (LG), 204-04

WHEELS AND TIRES

DESCRIPTION AND OPERATION

NOTES:

- This document describes the fault diagnosis to establish what the fault is with the TPMS system.
- The majority of TPMS faults, are not faults, but low pressure warnings, indicating the tires have lost air, and have reached the point where the TPMS ISO light has been illuminated.
- There is a TPMS diagnostic flow chart in TOPIx, to show a clear way of diagnosing a fault.

TPMS valve snaps

It is recommended that during the Vehicle Handover (or at subsequent Service intervals), advice should be given to customers that damage to the TPMS valve may occur if excessive side loading is applied during tyre inflation or checking.

TPMS valves are snapped/damaged when an excessive side load is applied as shown in the illustration E196994.

The damage occurs at the edge of the valve, as shown in the illustration E137619. If damage is on other area's of the valve, the warranty cost shouldn’t be covered by Jaguar Land Rover.

If available, the recommendation is to use a non 'wand' style inflator or the IMS compressor to maintain the pressure. This kit has a flexible style end piece.

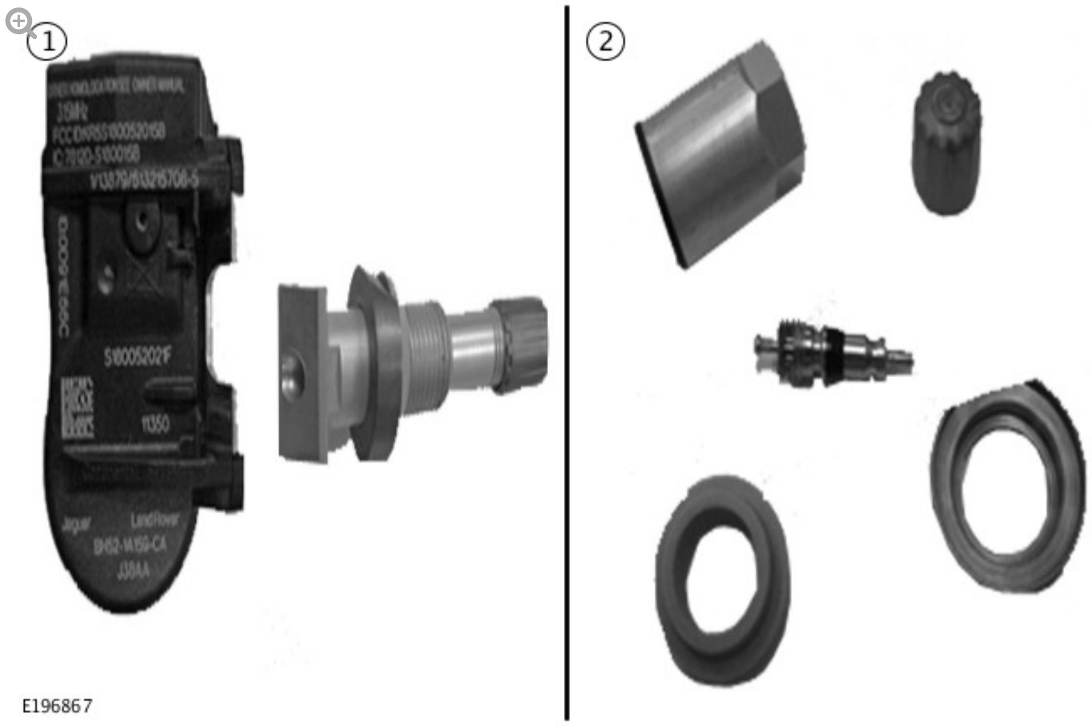
TG1C and TG1C valve stem

If the TPMS wheel unit is a TG1C, 2 part TPMS wheel unit, where the valve stem can be separated, then order just the LR043162 valve stem kit containing 5 replacement valve stems, nuts, washers. Use this sub assembly kit whenever a leak occurs between valve stem and the wheel.

NOTE:

> Please note that Jaguar Land Rover does not cover the snapped/damaged TPMS under warranty, this is not a manufacturing defect.

UK markets: Valve stem kit with a shorter valve stem is LR077043.



TPMS instrument cluster warnings

1.0 TPMS warning message on instrument cluster root cause: various causes, as follows:

Turn the ignition on and look at the Instrument Cluster.

- Does the TPMS ISO symbol flash for 72 seconds, and then stay on permanently: Yes go to **section 2.0**.
- Is the TPMS ISO Symbol illuminated permanently on the instrument cluster: Yes go to **section 1.1**.

If the TPMS ISO symbol does not illuminate: There is no fault or problem present on the vehicle at this time.

- Ask the customer if they have used a spare wheel fitted with a rubber valve, as this will cause a TPMS symptom fault to be displayed, this is expected behaviour, and there is no fault with the TPMS system. The use of a spare wheel is described in the vehicle handbook.
- Ask the customer if the instrument cluster displayed one or more wheels at 0 pressure. If yes then **section 1.2**.

- Ask the customer if they saw a warning TPMS message. If yes then **section 1.3**.
- Ask the customer if the TPMS fault message came on during a drive, and if the message cleared during a drive cycle. If the customer does indicate this happened, ask the customer if they have used additional accessories in the vehicle such as: USB chargers, insurance cameras, DC-DC converters, cool boxes, satellite navigation and radar detectors when the TPMS fault light came on. If the answer is yes, ask the customer to remove the items, and see if the TPMS fault light does not come on, during the next drive cycles.

1.1 Low pressure warnings.

When the TPMS ISO symbol is on all the time, or comes on during a journey, the TPMS system has detected that one or more tires are below the 80% requirement of the required pressure. The required pressure in the tire changes due to the temperature of the air inside the tire, being heated by the tire movement.

The instrument cluster will display a graphic of the vehicle showing which tire(s) are below the 80% of the required pressure. If only one wheel is low pressure then that wheel will be highlighted in yellow showing the actual pressure and the Recommended Cold Pressure (RCP) to inflate to.

The message to check all tires is because if the tire has naturally lost air, then the other three tires will also be close to the threshold to set the warning.

Using the instrument cluster menu, (ignition on), use the menu and arrow keys on the steering wheel to navigate to the TPMS menu, and select the pressure check option, and view the pressures of the tires, add air to the tyres as required. The correct pressure is shown in the pressure menu, on the instrument cluster. The recommended cold pressure is also shown on the tire label at the bottom of the B-pillar.

The instrument cluster should update as air is being placed into the tires. If the instrument cluster does not update, then drive the vehicle for 3 minutes above 15Mph. This will make sure the RF signals from the TPMS wheel units are received.

1.2 Instrument cluster displays one or more tires at 0 pressure without ISO TPMS symbol illuminated.

If the instrument cluster is showing one or more tires at 0 pressure, then inflate the tire by 300kPa and check the instrument cluster to see if the tire pressure has updated. If not then drive the vehicle for 8 minutes above 20Kph for the TPMS sensors to localise.

1.3 TPMS warning messages

1.3.1 Non EU or non NAS market TPMS warning messages.

In non EU or non NAS markets, when rubber valves are fitted, the TPMS instrument cluster will display the message **"Tire monitoring not available"**, after 10 minutes of driving. The instrument cluster message **"Tire monitoring not available"**, will come on every subsequent ignition cycle, until the TPMS sensors are re-fitted.

The instrument cluster message **"Tire Pressure Monitoring Available"**, will be displayed on the instrument cluster when all four TPMS wheel sensors have been detected, and will display for 20 seconds.

1.3.1 All market warning messages.

The instrument cluster message **"Tire Pressures too low for high speed"**, and the TPMS ISO symbol will be displayed when the tire pressures are 80% below the required pressure for high speed driving.

The instrument cluster and the tire pressure label at the bottom of the B-pillar on the driver's side of the vehicle will show the required pressures for high speed driving.

1.4 NAS market warning messages.

The load setting of the vehicle may be changed in the load setting menu, (vehicle dependant).

1.4.1 The instrument cluster message **"Tire Pressure set for Light Load"**, is a reminder that the load switching has been set to the light setting, and are the lower of the tire pressures recommended.

1.4.2 The instrument cluster message **"Tire Pressure set for Normal Load"**, is a reminder that the load switching has been set the normal setting, and is the highest tire pressures recommended.

1.5 EU and ROW warning messages.

The load setting of the vehicle may be changed in the load setting menu, (vehicle dependant).

1.5.1 The instrument cluster message **"Tire Pressure set for Light Load"**, is a reminder that the load switching has been set to the light setting, and are the lower of the tire pressures recommended.

1.5.2 The instrument cluster message **"Tire Pressure set for Heavy Load"**, is a reminder that the load switching has been set to the Heavy setting, and is the highest tire pressures recommended.

2.0 TPMS faults

Using the instrument cluster menu, with the ignition on and the vehicle stationary, does the instrument cluster show one or more wheels with a yellow cross highlighted above the wheel? No then go to **section 2.2**.

2.1 Wheel cannot be detected.

Check the wheel(s) to see if a rubber valve has been fitted. If it has replace with a TPMS wheel sensor. Jaguar Land Rover will not pay for the warranty.

If the wheel has a TPMS valve go to **section 2.2**.

2.2 TPMS fault messages.

Connect the approved Jaguar Land Rover diagnostic tool and read the DTCs from the TPMS wheel unit.

2.2.1 NAS markets only.

For NAS markets, please note the TPMS frequency has been set to 433 MHz from 16MY for all vehicles.

2.2.1 EU markets.

EU markets the TPMS system is always 433 MHz.

2.2.2 Wheel unit signals not received from start of a drive cycle.

- C1D21-05 (Any wheel unit - Failed reception from beginning of TPMS Drive cycle)
- C1A56-31 (Left Front Tire Pressure Sensor and Transmitter Assembly - Mute from start of drive cycle)
- C1A58-31 (Right Front Tire Pressure Sensor and Transmitter Assembly - Mute from start of drive cycle)
- C1A60-31 (Left Rear Tire Pressure Sensor and Transmitter Assembly - Mute from start of drive cycle)
- C1A62-31 (Right Rear Tire Pressure Sensor and Transmitter Assembly - Mute from start of drive cycle)

Possible Causes:

1. Rubber valve fitted go to **section 2.1**.
2. 315 MHz TPMS wheel unit fitted on a 433 MHz vehicle. (NAS markets Inspect the wheel unit). If a 315 MHz wheel unit is fitted, Jaguar Land Rover will not pay for a replacement.
3. TPMS wheel unit in ship mode, turned off or customer purchased part. Using a handheld LF tool, turn the TPMS wheel unit on. Jaguar Land Rover will not pay for this as customer purchased.
    - If no TPMS LF hand tool available, then change the TPMS wheel unit to a new one. Jaguar Land Rover will not pay for a replacement, as part not purchased through approved process.
4. Non Compatible TPMS wheel unit fitted. Inspect wheel unit and check Jaguar Land Rover part number. Loc Sync systems will only work with

FW93 1A159 AA/AB or GX63 1A159 AA.

5 The TPMS wheel unit battery has failed or the wheel unit has been damaged when tire was replaced.

Replace the identified wheel unit.

NOTE:

> Please note that Jaguar Land Rover does not cover these cost under warranty, this is not a manufacturing defect.

2.2.3 Wheel units not recognised during a drive cycle.

- C1A56-93 (Left Front Tire Pressure Sensor and Transmitter assembly no operation (During TPMS drive cycle)
- C1A58-93 (Right Front Tire Pressure Sensor and Transmitter assembly no operation (During TPMS drive cycle)
- C1A60-93 (Left rear Tire Pressure Sensor and Transmitter assembly no operation (During TPMS drive cycle)
- C1A62-93 (Right rear Tire Pressure Sensor and Transmitter assembly no operation (During TPMS drive cycle)

Possible Causes:

1. One or more wheels swapped over in less than 15 minutes (winter wheel warehouse replacement).
2. RF interference of the TPMS signal from the wheel units to the receiver.
3. TPMS wheel unit damaged.
4. Battery failing, (wheel unit over 6 years old).

Go to **section 3.0**.

2.2.4 Wheel unit localisation.

- C1D18-00 Wheel localisation failed.

Check the ABS system for DTCs set. The TPMS system uses the ABS wheel sensor signals to localise the wheel units to the correct wheel location.

The TPMS system needs to receive the RF transmissions from all four wheel units, to enable the location of the TPMS wheel unit and the TPMS wheel unit ID to be matched. If a rubber valve has been fitted to a wheel, or the TPMS sensor is not transmitting, on the fifth journey and greater than 10 minutes, a localisation DTC will be set.

Make sure all four TPMS wheel units are working using the approved Jagur Land Rover diagnostic tool deflation test.

If there is RF interference, this can cause the localisation to fail.

Go to **section 3.0**.

2.2.5 Incorrect TPMS wheel unit installed.

- C1A56-4A (Left Front Tire Pressure Sensor and Transmitter assembly - Incorrect component installed)
- C1A58-4A (Right Front Tire Pressure Sensor and Transmitter assembly - Incorrect component installed)
- C1A60-4A (Left rear Tire Pressure Sensor and Transmitter assembly -

Incorrect component installed)

- C1A62-4A (Right rear Tire Pressure Sensor and Transmitter assembly - Incorrect component installed)

The vehicle has been fitted with TPMS TG1B wheel sensors that are not compatible with the TPMS system fitted on the vehicle.

New TPMS TG1C wheel sensors are required to be fitted. Jaguar Land Rover will not pay for this warranty.

2.2.6 Low pressure sensor fitted on a high pressure system. Incorrect wheel sensor fitted.

- C1A56-85 (Left Front Tire Pressure Sensor and Transmitter assembly – Low pressure sensor fitted)
- C1A58-85 (Right Front Tire Pressure Sensor and Transmitter assembly - Low pressure sensor fitted)
- C1A60-85 (Left rear Tire Pressure Sensor and Transmitter assembly - Low pressure sensor fitted)
- C1A62-85 (Right rear Tire Pressure Sensor and Transmitter assembly - Low pressure sensor fitted)

If any of these DTCs are noted, then the tire pressure sensor needs to be changed to a High Pressure wheel unit LR070840.

2.2.7 ECU internal failure.

- B1182-96 – ECU internal failure – Replace the ECU

2.2.8 CAN communication bus missing message.

- U0001-87 CAN Communication missing message

One or more CAN frames have not been received from the PCM, ABS, BCM, IPC or GWM. Check the CAN network, and other ECUs DTCs to work out which ECU has got the fault.

2.2.9 Invalid CAN signal received.

- U0001-81 Invalid CAN signal received

The signals received are vehicle speed from ABS, ambient temperature from PCM and ambient air pressure from the PCM. Read local snap shot data 0xD020 and the 4 byte code will identify the ECU at fault.

2.2.10 Implausible CAN signal received.

- U0001-86 Implausible CAN signal received from ABS, PCM, BCM, IPC or GWM

Read local snap shot data 0xD020 and the 4 byte code will identify the ECU at fault.

2.2.11 CAN Bus off.

- U0001-88 CAN bus off

Check the CAN bus Voltages.

2.2.12 CAN configuration errors.

No CAN master configuration ID received.

Flash latest software in to TPMS Module.

2.2.13 Missing local configuration file.

- U3000-51 Missing Local Configuration File

Perform a software update on the TPMS module selecting the correct LCF file.

2.3 Car configuration problems.

- U2300-54 Central Configuration – missing calibration
- U2300-55 Central configuration – not configured

- U2300-56 Central configuration invalid / incompatible

Perform a software update on the GWM, and check the configuration for the vehicle is correct.

These DTCs can occur when the wheel or tire size is not compatible with the vehicle.

3.0 Wheel unit investigation

Ask the customer if they have used additional accessories in the vehicle such as: USB chargers, insurance cameras, DC-DC converters, cool boxes, satellite navigation and radar detectors when the TPMS fault light came on. If yes, ask the customer to remove the items, and see if the TPMS fault light does not come on.

Break the bead and inspect the TPMS wheel unit and compare the wheel unit ID, with the wheel ID as read by the approved Jaguar Land Rover diagnostic tool. If the TPMS wheel IDs are different, programme the new wheel ID into the approved Jaguar Land Rover diagnostic tool, and run the deflation test. Make sure the TPMS wheel unit is working.

If the TPMS wheel unit is not detected with the new wheel unit, then ask the customer where the TPMS wheel unit was purchased. The wheel unit is in ship mode, and needs to be put into park mode with a LF tool. Jaguar Land Rover will not pay the warranty for this claim, this is not a manufacturing defect.